

Rowan Torbitzky-Lane

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Education

B.S. Mathematics - GPA: 4.00 Lindenwood University

Expected: May 2027

B.S. Computer Science - GPA: 3.80 Missouri University of Science and Technology

Graduated: December 2024

Skills

- | | | |
|----------|---------|---------|
| • Python | • Unity | • CUDA |
| • Rust | • SQL | • Godot |
| • Docker | • Linux | • Nix |

Relevant Coursework

- | | | |
|----------------------|------------------------------|-----------------------|
| • Algorithms | • Differential Equations | • Linear Algebra I/II |
| • Data Structures | • Calculus III | • Physics I/II |
| • Introduction to AI | • Probability and Statistics | • Linear Regression |

Experience

Computer Graphics Intern - USC Institute for Creative Technologies May 2026 - August 2026

- Accelerate mesh/splat correspondence calculation to under one second via C# to Rust FFI
- Train Gaussian Splats on bounded and unbounded scenes
- Extract single objects from Gaussian Splats
- Utilize Nix for programming environment setup

NSF Research Assistant - Missouri University of Science and Technology May 2024 - October 2024

- Train novel, bi-directional LLMs on Genome Sequences to correct errors in reads
- Clean and analyze textual data with Scikit-learn and Scikit-bio
- Incorporate huggingface library for model testing

Research Assistant - Center for Intelligent Infrastructure May 2023 - October 2023

- Calculate optical flow with MATLAB and C++ on campus HPC via Slurm
- Coordinate Web Requests with Django to interact with internal PostgreSQL server
- Improve frontend with Typescript, such as adding video upload capability
- Extract image features with OpenCV, using optical flow to predict annotations in subsequent video frames
- Perform Annotations for image segmentation Convolutional Neural Networks

Data Science Co-Op - Hunter Engineering Company January 2022 - August 2022

- Utilize YOLO and PyTorch in license plate recognition problems
- Pioneer object detection models to eradicate bad data before annotators see it
- Innovate data pipeline by automating our data downloading process
- Design and containerize Python APIs with FastAPI and Flask within Docker and begin creating endpoints within Azure
- Leverage Pandas and Microsoft SQL in data processing pipelines
- Enforce consistent code style with automated formatting checks in internal Azure DevOps pipeline

Introduction to CS Grader - Missouri University of Science and Technology January 2021 - May 2021

- Author automated Python test cases validating student submissions against assignment specifications
- Translate test failures into actionable feedback for students new to programming

Personal Projects

Self-Hosted NixOS Server - SQL, YOLO, Docker, FirewallD, Git, Nginx, CUDA January 2023 - Present

- Automate flake updates daily via CI/CD pipeline on GitLab
- Train bird object detection Convolutional Neural Networks
- Host game servers via Docker

Rust Genetic Programming Framework - Polars, rust_decimal March 2025 - Present

- Perform pre-processing on Time Series Data
- Utilize test driven development philosophy involving writing unit tests
- Write declarative and procedural macros for reduced boilerplate
- Utilize rust_decimal's floating point precision for increased accuracy

Talks & Events

Talk: Genetic Programming in Python @ PySTL https://www.youtube.com/watch?v=ZzYi_j2IYuY September 18th, 2025

Talk: Actix Web Framework @ STL Rust <https://rumble.com/v72mid6-actix-web-unleashed.html> December 4th, 2025

Hackathon: SIGGRAPH 2026 Worlds in Action Hack @ MG Studio luma.com/u7pmhw92 July 18th, 2026